

Heights Of Lunar Mountains

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Abstract

The aim of this experiment was to measure the height of mountains formed by craters on the lunar surface. A travelling microscope and a negative photo of the moon were utilized to do this. The selected craters shadow lengths in the photo were measured, and after some geometric manipulation involving approximating the terminator of the moon as an ellipse, the heights of the mountains were obtained using trigonometry. To access the detailed results quickly the reader can go to the page 7 and search for Figure 5. The heights of the mountains ranged from 5 km to 2.5 km with uncertainties ranging from 0.5 km to 1.25 km. The main source of errors in this experiment was the uncertainty in the minor axis of the ellipse.

1. Introduction

Before the Space Race between USA and USSR, The topological knowledge of the moon's surface was rather limited. This state of affairs changed quickly as in the early 60s an international programme, aiming to measure and document moon's surface accurately, was launched [1]. In the process, many detailed photos of the moon were produced and analysed. The photo used in this experiment was taken by a refracting high-altitude telescope in France. The telescope itself is rather rare as it has a focal length of 60 foot (18.3 meters) and a 2-foot achromatic objective lens. The image was recorded on a photographic film in a quick 1-second exposure. [2]

2. Theory

2.1 Half-Moon Case

The moon was treated as a perfect sphere, The geometry of the moon was first analysed in the case of the image being a half moon, with the terminator as a straight line.

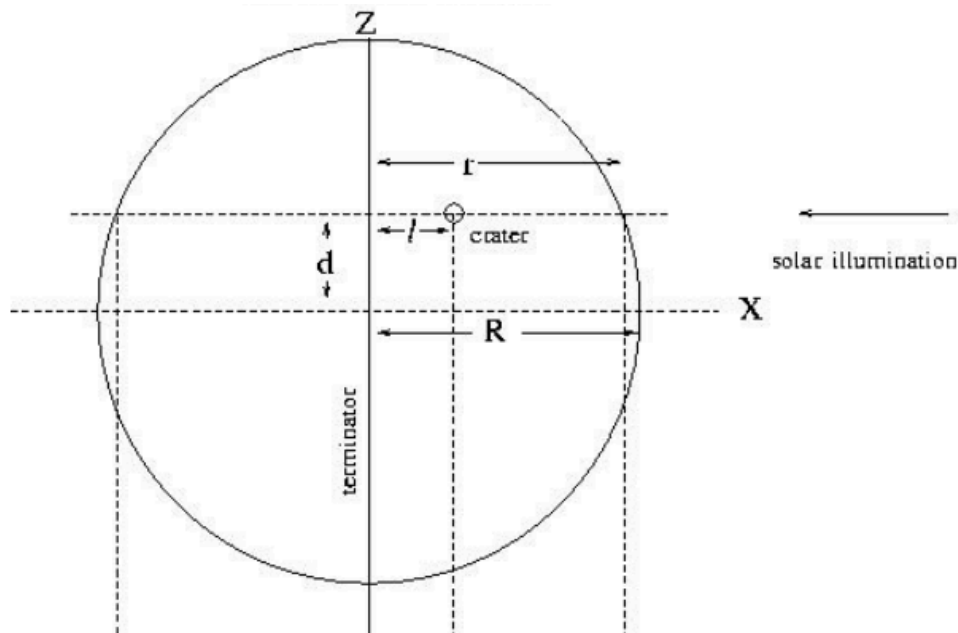


Fig 1: Image of a half moon with a mountainous crater, the terminator is the line at which the moon turns from light to dark, r is the perpendicular distance from the terminator to the edge of the moon, d is the vertical distance from the centre of the moon to the measured crater, l is the perpendicular distance from the terminator to the measured crater and R is the radius of the moon. Reproduced from [2]

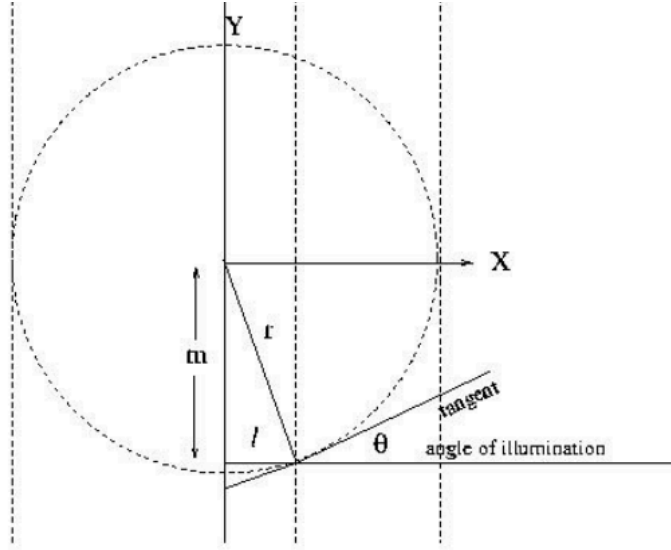


Fig 2: Top view of the moon shown in fig 1 with θ as the angle of illumination, m is the distance from the pole to the measured crater and θ is the angle between the tangent to the moon's surface and the direction of solar illumination. Reproduced from [2].

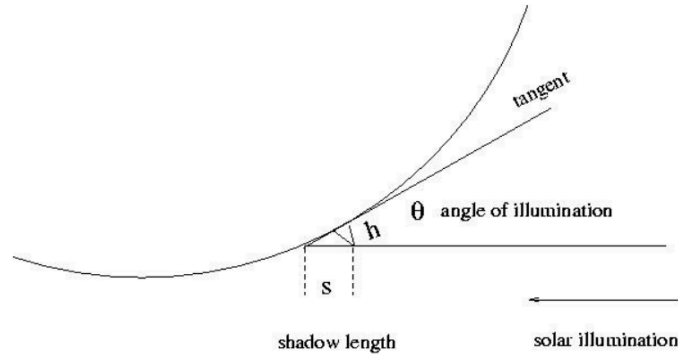


Fig 3: shows a mountain on the lunar surface, h is the mountain height, s is the length of the mountains shadow and θ is the angle of the illumination. Reproduced from [2].

Using Pythagoras' theorem $d^2 + r^2 = R^2$ and $m^2 + l^2 = r^2$. R , d and l can be measured using the travelling microscope, therefore m can be determined.

As shown in figure 2, $Y = -m = -(r^2 - l^2)^{1/2}$ and $X = l$.

As $X^2 + Y^2 = r^2$ for a circle with radius r , taking the derivative gives $2XdX + 2YdY = 0$.

Therefore for the tangent at XY ,

$$\tan(\theta) = \frac{dy}{dx} = \frac{-X}{Y} = \frac{l}{\sqrt{r^2 - l^2}} \quad (1)$$

The mountain height h can then be determined using the equation,

$$\sin(\theta) = \frac{h}{s} \quad (2)$$

2.2 Adjustment for small rotation away from the half moon

In the image analysed in this experiment the moon had rotated a small angle ϕ away from the half moon, meaning the theory had to be adjusted to take this into consideration. The terminator of the moon now looks like an ellipse as shown in figure 4.

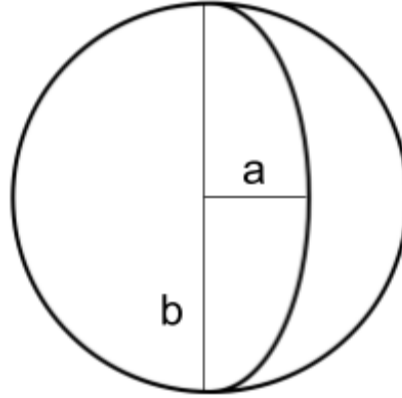


Fig 4: The moon after rotating an angle ϕ , the terminator is now an ellipse. Reproduced from [2].

The equation of an ellipse is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad (3)$$

Where a is the semi minor axis and b is the semi major axis of the ellipse.

This can also be written as:

$$\frac{a^2}{b^2} = \frac{y_2^2 - y_1^2}{x_1^2 - x_2^2}, \quad (4)$$

meaning the values of a and b can be obtained by measuring the coordinates of several points on the ellipse using the travelling microscope and taking the mean of these measurements.. These values of a and b can then be used to correct the distance from the terminator to the ellipse, l , by subtracting a correction c from the measured crater's x coordinate, so $l = x - c$, where

$$c = \sqrt{b^2 - y^2 \frac{b^2}{a^2}} \quad (5)$$

3. Experimental details

The experimental setup is shown in figure 5. It consists of a travelling microscope and the photo of a moon near a half-moon phase. The first part of the experiment consisted of setting up a coordinate system with origin on the centre of the moon, marked with a red dot. This position was recorded and all distances were calculated in relation to this origin. From this point onwards, the photography was not moved or rotated. The second part of the experiment revolved around determining the major and minor axis of the ellipse used to approximate the terminator. This was done by measuring the major axis (this is just the diameter of the moon) followed by taking a range of measurements of the points on the terminator using a travelling microscope. These points were then used to determine a minor axis of the ellipse. This part of the experiment was essential as it reduced the main systematic source of errors in this experiment. A third part of the experiment consisted of picking a lunar object from an atlas of the lunar surface [3], locating its position in the photograph and measuring the length of its shadow using the travelling microscope. In order to minimize another source of systematic errors, the blur of a photo, shadow length was measured to the grey area instead of measuring to the dark or light part of the shadow, as this would minimise the deviation from the true length of the shadow. For each shadow, 3 measurements were taken to reduce the random error. In total heights of 8 objects were calculated. For the ellipse fitting, 10 points were measured.

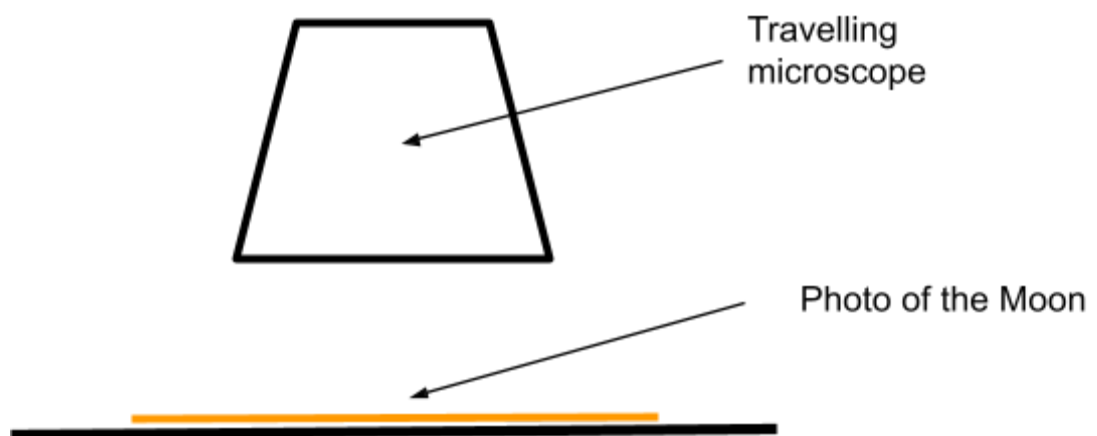


Fig 5. Schematic diagram of the experiment

4. Results

4.1 Calculating the Ellipse

The moon was assumed to be a sphere, with radius 1737 km. The radius of the moon measured using the travelling microscope was 76.18 ± 0.005 units, giving a conversion factor of 22.8km per unit length of the microscopes measuring device. The terminator of the moon was approximated by an ellipse. The coordinates of ten points on the ellipse were taken and l and d for each point were calculated by taking the absolute value of the distance from the centre of the moon. These 10 points were then subbed into equation 3 above to give values for a and b , the semi minor and major axis. The mean of these values was taken, giving $a = 16.80 \pm 2.23$, $b = 76.18 \pm 0.005$ and $b/a = 4.53$. Using these values l can be corrected for any measured x and y coordinates. The uncertainties in the measurements with travelling microscope were 0.005 in the arbitrary units, half the precision of the instrument. The uncertainty in the minor axis of the ellipse was calculated to be 2.23 arbitrary units, which as a percentage is 13.3%, a fairly large error.

4.2 Measuring Craters

The interior shadow length of eight craters and their x and y coordinates was then measured, shown in table 2 below:

Crater Name	Interior shadow length Arbitrary units +/-0.005	X coordinate Arbitrary units +/-0.005	Y coordinate Arbitrary units +/-0.005
Barocius	0.718	108.876	72.332
Werner	1.33	112.493	88.361
Purbach	0.67	110.081	83.850
Gemma Frisius	1.87	111.670	104.020
Sacrobosco	1.20	109.665	105.192
Abulfeda	0.93	103.713	103.930
Menelaus	0.87	105.230	166.2777
Archimedes	0.12	114.182	207.872

Table 1. Coordinates and shadow lengths of observed objects

Each X coordinate was corrected by c , calculated according to equation 5. The corrected l values were then subbed into equation 1 to give a value of θ , which was then used in equation 2 along with the shadow length to give a value for the height of the measured objects, as shown in graph 1 below

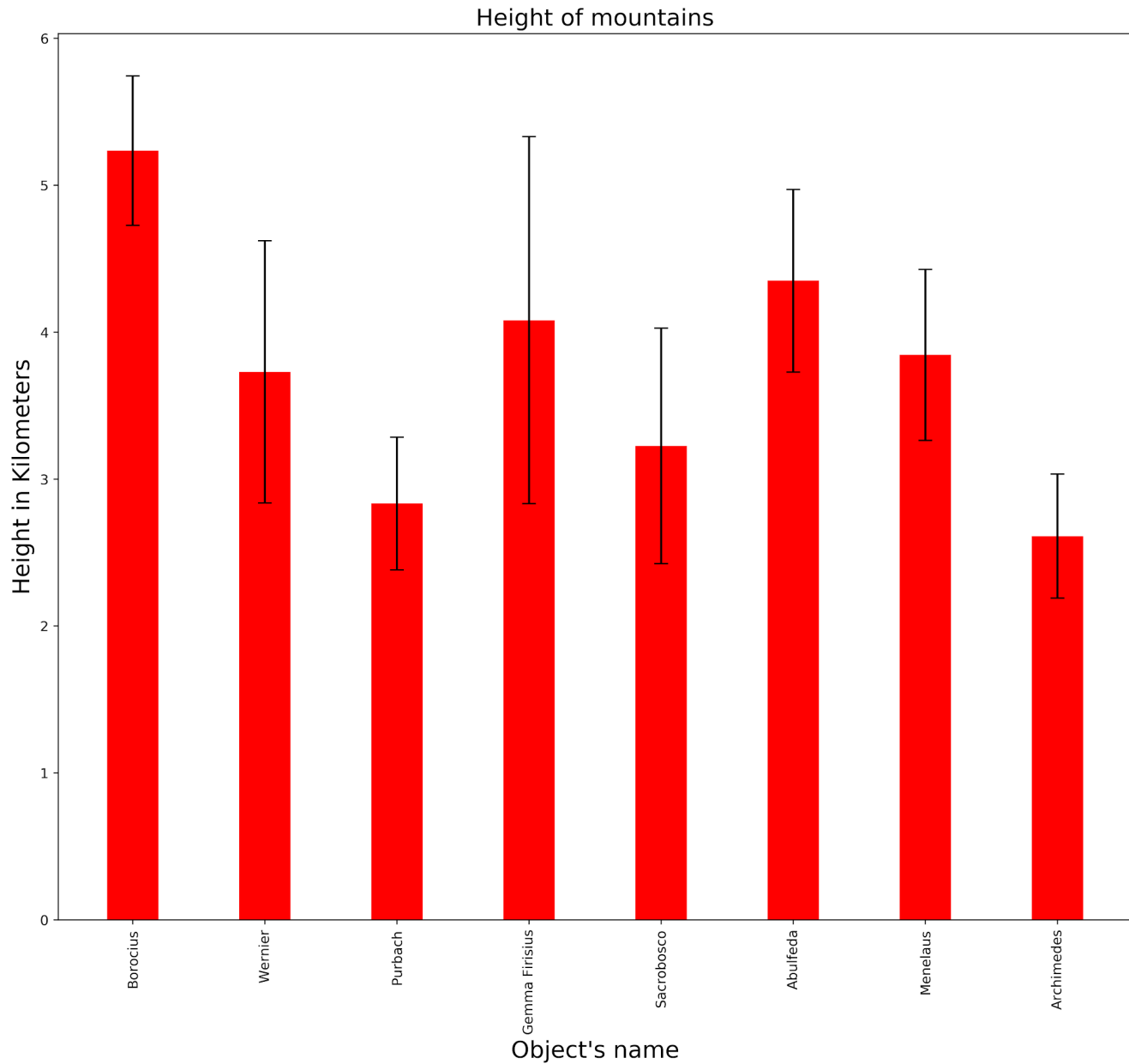


Fig 5 - bar chart showing the heights of measured objects

5. Conclusion

In summary all results were within 3 standard deviations of the accepted heights for each object. However, all of the results were significantly below these accepted heights. This could be because of a systematic error in the measurement of the minor axis of the ellipse. The point at which the terminator changed from light to dark was fairly arbitrarily determined which could lead an error in the value of the ellipse. Overall random errors were insignificant compared to the systematic errors present in the experiment. The experiment and analysis did not account for the blur of the photo or any atmospheric effects. Another unaccounted source of errors was the ambiguity in determining whether the observed point laid on the terminator or not. This might have been improved by performing digital analysis of a photo and defining a border in terms of a certain level of light.

References

1. Wieczorek, M., "The Constitution and Structure of the Lunar Interior". *Reviews in Mineralogy and Geochemistry*, 60, 1, 221-364, 2006.
2. Meaburn, J., "Heights of Lunar Mountains", 2nd yr lab script, School of Physics and Astronomy, University of Manchester, 2005.
3. Wood, C. and Collins, M., *21st century atlas of the Moon*, West Virginia University Press, 1st edition, 2012.